MeshSat Compact v2 (model MSC-2): instructions and safety information

Read this before first use and keep it with the unit.

What it is

MeshSat Compact v2 is a portable messaging gateway. It carries text messages between nearby Meshtastic LoRa handhelds, your phone (over Bluetooth, with the MeshSat app for Android or iOS) and the Iridium satellite network. It is not an emergency or distress device. Do not rely on it to call for help.

Before first use

1. Screw both antennas onto their sockets on the panel first: the short helical satellite antenna on the socket marked SAT, the LoRa antenna on the socket marked LORA. Never power the unit without both antennas fitted.
2. Fit one 18650 Li-ion cell as described under "Battery and charging" in the holder, observing the + and - marks. Do not use a damaged, swollen or dented cell.
3. Close the lid with all eight screws until it touches the rim. Do not overtighten.
4. Charge through the USB-C socket until the display shows a full battery.

Pairing with your phone

- Open the MeshSat app and pick the unit from the Bluetooth list. The unit shows a 6-digit PIN on its display; type it into the app.
- The PIN changes at every pairing. Never tell it to anyone.
- After 3 failed pairings in a row, the unit refuses pairing for 5 minutes, and longer after each further failure.

Radio region

The unit is sold for use in the European Union and ships set to the EU 868 MHz region. In the EU use the EU 868 MHz setting only. Outside the EU, local rules decide whether and how you may use it.

Satellite: arm before sending

The satellite modem is switched off unless you arm it. While it is off, the unit keeps satellite messages in a queue and you may wear the unit on your belt or strap.

1. Take the unit off your body. Set it down or hang it up with the satellite antenna pointing straight up, with a clear view of the sky, at least 1 m above the ground and away from metal surfaces.
2. **Keep it at least 30 cm away from people while it is armed.**

3. Arm it: hold the panel button for 3 seconds, or tap Arm satellite in the app. The display shows SAT ARMED.
4. The queued messages go out; this can take from seconds to several minutes, depending on the sky view. Messages for you arrive while it is armed.
5. The unit disarms itself 15 minutes after the last satellite traffic, 30 minutes after arming at the latest, when the cell runs low, and at every restart. You can disarm it in the app. While it shuts the modem down, SAT ARMED blinks. **Only put it back on your body when the display shows SAFE TO WEAR.**

While the satellite modem transmits, the LoRa radio pauses. Sending by satellite needs an active Iridium service plan on the modem.

Quick message button

A short press puts your preset message for your preset contact in the queue. Set both in the app. The message goes out by satellite once you arm the unit. It is a convenience shortcut, not an emergency service: nobody monitors these messages for you.

Interfaces and services

- **Bluetooth:** the MeshSat app's connection (settings, messages, positions, the node list, the security log), the battery level and the app's satellite channel. Only a paired phone can use it.
- **LoRa (868 MHz):** Meshtastic mesh messages, positions and node information.
- **Satellite (Iridium):** messages to and from your satellite service, only while armed.
- **USB-C:** charging and setup. Software updates come only through the app.
- **GNSS receiver:** the unit determines its own position. It shares it only if you turn position sharing on.
- **WiFi:** switched off.

Your data and privacy

- The unit stores your settings, its own security keys, the list of nearby units, recent messages and its position.
- Sharing your position is off until you turn it on in the app.
- The default LoRa channel is a public channel: anyone with a Meshtastic radio nearby can read it. For private messages, create your own channel in the app, which gives it its own key, or send direct messages.
- Texts on the satellite channel are forwarded to the satellite. Give that channel its own key in the app and share it only with your team: anyone who has the key can use your satellite allowance (the unit sends at most 10 satellite sessions a day, and suppresses duplicate copies of a message for 24 hours). The unit does not forward anything while the satellite channel has no key of its own.

- Messages you send from the MeshSat app can be encrypted end to end before they reach the satellite: turn encryption on in the app's settings. **Messages the unit forwards by satellite on its own, from mesh handhelds without the app, are not encrypted.** Send private content from the app with encryption on.
- The unit keeps a security log (pairings, setting changes, updates, resets, arming) with times, without message content or positions. You can read it in the app.
- To erase everything (settings, keys, Bluetooth pairings, messages, node list, security log), use Factory reset in the app.

Software updates

Updates are published at meshsat.net. The unit never updates itself: you start every update from the app. The unit accepts only updates signed by MeshSat and never an older version. Keep the unit up to date.

Battery and charging

- The unit comes with a Samsung INR18650-35E cell (3,6 V, 3 480 mAh, 12,5 Wh). A replacement must be a genuine 18650 Li-ion cell from a known maker (such as LG, Samsung, Sony/Murata or Panasonic): flat top, 4,2 V full charge, rated for charging at 0,5 A between 0 and 45 °C.
- To replace the cell: switch the unit off and unplug the charger, undo the eight lid screws, lift the old cell out of its holder, fit the new one observing the + and - marks, and close the lid with all eight screws until it touches the rim. Do not overtighten. Take the old cell to a battery collection point.
- Charge only through the USB-C socket from a 5 V USB charger. Charge between 10 and 35 °C; outside that the unit pauses charging. Use between -10 and 45 °C.
- There is no temperature sensor on the cell. The unit reads the temperature of the T-Beam's charger chip and, while the satellite modem runs, of the modem's board. It charges only while the charger chip's reading, corrected for its own heat, is between 10 and 35 °C and the modem's board is at 10 °C or more and not over 50 °C. After a reading under 10 °C charging waits 60 minutes, and after one over 35 °C it waits 30 minutes. For the first 10 minutes after switching on it does not charge. Before the firmware has started, the charger runs on its own.
- Do not charge the unit unattended inside a closed bag, a sleeping bag or a vehicle in the sun.
- If the case gets hot, smells or swells: stop using it, unplug the charger, put the unit on a non-flammable surface away from people and anything that burns, and let it cool completely before touching it again. Do not open it while it is hot.
- Do not open, crush, short-circuit or heat the cell. Do not leave it in the unit for months without use; charge it now and then.
- When the cell runs low, the unit shuts the satellite modem down first and will not arm it again until the cell is recharged.

Water and handling

The lid closes on a silicone O-ring and the panel parts have their own seals, but the case has no tested water or dust rating. Do not submerge it. Keep the lid closed and the USB-C cap fitted outdoors. Do not drop the unit onto the antennas.

Substances

Brass parts of the unit (threaded inserts, antenna connectors) may contain lead (CAS 7439-92-1) above 0,1 % by weight (REACH Regulation, Article 33). Do not file, grind or melt these parts.

Radio specifications

Radio	Frequency band	Maximum power transmitted
LoRa (Meshtastic)	869,4 to 869,65 MHz (EU 868 setting)	22 dBm (158 mW) at the antenna socket
Bluetooth Low Energy	2400 to 2483,5 MHz	-3 dBm (0,5 mW)
Iridium satellite, only while armed	1616 to 1626,5 MHz	32 dBm (1,58 W) average, 37 dBm (5 W) peak, at the antenna socket
GNSS (position)	1559 to 1610 MHz	receive only

Disposal

Do not put the unit or the cell in household waste. Remove the cell and take both to an electronics and battery collection point.

Declaration of conformity

Hereby, Elli.Z.G., trading as MeshSat, declares that the radio equipment type MSC-2 is in compliance with Directive 2014/53/EU. The full text of the EU declaration of conformity is included with this unit.

Manufacturer

Elli.Z.G., trading as MeshSat
Floresstraat 17, 2315 HP Leiden, the Netherlands
hello@meshsat.net | meshsat.net

Model MSC-2. The serial number is on the label on the back of the case.